The L^∞ condition is redundant

A completion of the compact $T(1)$ theorem à la Stein

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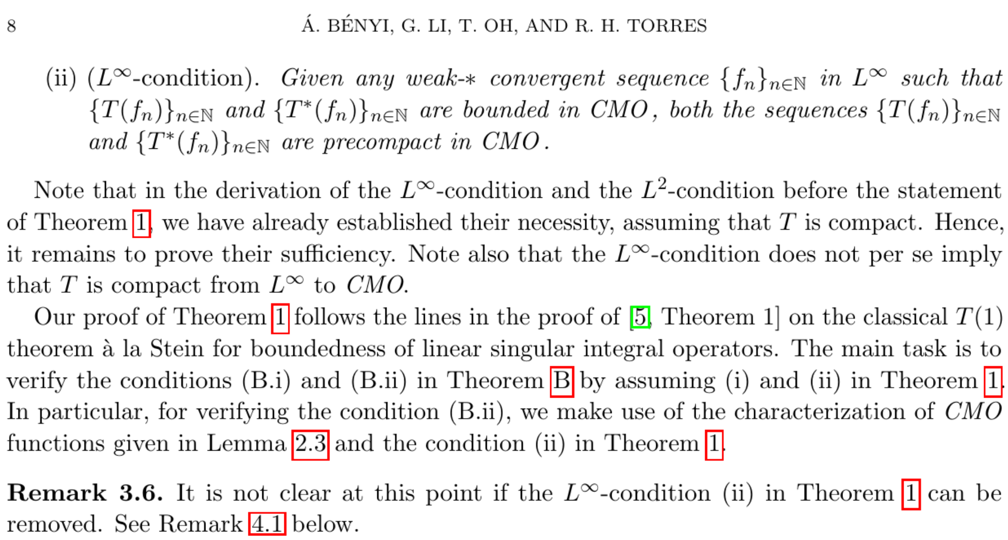
Result. Condition (i), the normalized-bump L^2 decay, is by itself necessary and sufficient in Bényi–Li–Oh–Torres's compact $T(1)$ theorem. Their L^∞ condition (ii) is automatic.

1 Source question and answer

Bényi, Li, Oh, and Torres prove a compact $T(1)$ theorem à la Stein for a linear singular integral operator $T : \mathcal{S}(\mathbb{R}^d) \to \mathcal{S}'(\mathbb{R}^d)$ with a standard Calderón–Zygmund kernel [1, Theorem 1]. Their condition (i) says that, for some bump order M and every normalized bump ϕ of that order,

$$R^{-d/2}\big(\|T(\phi^{x,R})\|_2 + \|T^*(\phi^{x,R})\|_2\big) \longrightarrow 0 \quad \text{as } |x| + R + R^{-1} \longrightarrow \infty, \tag{1}$$

where $\phi^{x,R}(y) = \phi((y-x)/R)$. Their condition (ii) is a precompactness requirement in CMO for images of weak-star convergent L^∞ sequences. Remark 3.6 asks whether (ii) can be removed.



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(ii) (L^∞-condition). *Given any weak-$*$ convergent sequence $\{f_n\}_{n\in\mathbb{N}}$ in L^∞ such that $\{T(f_n)\}_{n\in\mathbb{N}}$ and $\{T^*(f_n)\}_{n\in\mathbb{N}}$ are bounded in CMO, both the sequences $\{T(f_n)\}_{n\in\mathbb{N}}$ and $\{T^*(f_n)\}_{n\in\mathbb{N}}$ are precompact in CMO.*

Note that in the derivation of the L^∞-condition and the L^2-condition before the statement of Theorem 1, we have already established their necessity, assuming that T is compact. Hence, it remains to prove their sufficiency. Note also that the L^∞-condition does not per se imply that T is compact from L^∞ to *CMO*.

Our proof of Theorem 1 follows the lines in the proof of [5, Theorem 1] on the classical $T(1)$ theorem à la Stein for boundedness of linear singular integral operators. The main task is to verify the conditions (B.i) and (B.ii) in Theorem B by assuming (i) and (ii) in Theorem 1. In particular, for verifying the condition (B.ii), we make use of the characterization of *CMO* functions given in Lemma 2.3 and the condition (ii) in Theorem 1.

Remark 3.6. It is not clear at this point if the L^∞-condition (ii) in Theorem 1 can be removed. See Remark 4.1 below.

Figure 1: The source question, arXiv:2405.08416v2, PDF page 8.

2 Proof intuition

The source proof fixes a large cutoff ϕ_R and proves $T(\phi_R) \in \text{CMO}$, but some of its estimates are not uniform as $R \to \infty$; condition (ii) is then used to extract a strongly convergent subsequence. The key is to avoid a global cutoff.

For each test cube Q, cut off the constant function at a scale $A\ell(Q)$ centered on Q. The local part is controlled directly by the normalized-bump decay, while kernel Hölder regularity makes the far part nearly constant on Q. Quantitatively,

$$\mathrm{MO}_Q(T(1)) \lesssim A^{d/2}\varepsilon_T(x_Q, A\ell(Q)) + A^{-\delta}.$$

Letting A tend to infinity sufficiently slowly makes both terms vanish, uniformly for small cubes, large cubes, and fixed cubes translated to infinity. Thus $T(1), T^*(1) \in \mathrm{CMO}$ without condition (ii).

Theorem 1. *Let T be a linear singular integral operator with a standard Calderón–Zygmund kernel, as in Theorem 1 of [1]. Then T extends compactly to $L^2(\mathbb{R}^d)$ if and only if* (1) *holds. In particular, condition (ii) of that theorem is redundant.*

3 The localized cutoff estimate

For a locally integrable f, write

$$\mathrm{MO}_Q(f) = \frac{1}{|Q|}\int_Q |f - f_Q|, \qquad f_Q = \frac{1}{|Q|}\int_Q f.$$

Choose once and for all $\chi \in C_c^\infty(B(0,1))$ equal to one on $B(0,1/2)$. Multiplying χ by a fixed constant makes it a normalized bump of order M; by linearity, (1) therefore applies to χ itself. Set

$$\varepsilon_T(x,R) = R^{-d/2}\|T(\chi^{x,R})\|_2, \qquad \varepsilon_{T^*}(x,R) = R^{-d/2}\|T^*(\chi^{x,R})\|_2.$$

Lemma 2. *Let Q have center x_Q and side length ℓ. If $A \geq A_0(d)$, then*

$$\mathrm{MO}_Q(T(1)) \lesssim A^{d/2}\varepsilon_T(x_Q, A\ell) + A^{-\delta}, \tag{2}$$

where δ is the Hölder exponent of the kernel. The analogous estimate holds for $T^*(1)$.

Proof. Put $\eta_Q = \chi^{x_Q, A\ell}$. For A_0 large enough, $\eta_Q = 1$ on a fixed dilate of Q. The standard distributional definition of $T(1)$ shows that on Q, modulo an additive constant,

$$T(1)(x) = T(\eta_Q)(x) + F_Q(x), \tag{3}$$

where one may take

$$F_Q(x) = \int_{\mathbb{R}^d} [K(x,y) - K(x_Q,y)](1 - \eta_Q(y))\,dy.$$

This integral is absolutely convergent: its integrand is supported where $|y - x_Q| \gtrsim A\ell$, and the kernel difference decays like $|x - x_Q|^\delta |y - x_Q|^{-d-\delta}$. Consequently,

$$\sup_{x\in Q} |F_Q(x)| \lesssim A^{-\delta}.$$

For the local part, Cauchy–Schwarz gives

$$\frac{1}{|Q|}\int_Q |T(\eta_Q)| \leq |Q|^{-1/2}\|T(\eta_Q)\|_2 \lesssim A^{d/2}\varepsilon_T(x_Q, A\ell).$$

Subtracting averages in (3) proves (2). Replacing K by its transposed kernel gives the assertion for T^*. □

The representation (3) can equivalently be verified against mean-zero test functions supported in Q; subtracting $K(x_Q, y)$ makes the far-field integral absolutely convergent and removes the irrelevant additive constant.

4 Condition (i) forces the two CMO conditions

Define the decreasing tail modulus

$$\Omega(s) = \sup_{|x|+R+R^{-1}\geq s} \big(\varepsilon_T(x,R) + \varepsilon_{T^*}(x,R)\big).$$

For the fixed cutoff χ, condition (1) says precisely that $\Omega(s) \to 0$. Put $\omega(s) = \max\{\Omega(s), s^{-1}\}$ and

$$\alpha = 1 - \frac{d}{2(d+1)} > 0.$$

We verify the three vanishing-oscillation conditions in the Bourdaud–Uchiyama characterization quoted as Lemma 2.3 in [1].

Small cubes. Let $u = \ell^{-1}$ and

$$A(u) = \min\{u^{1/2}, \omega(u^{1/2})^{-1/(d+1)}\}.$$

Then $A(u) \to \infty$, while $A(u)\ell \leq u^{-1/2} \to 0$. Hence $(A\ell)^{-1} \geq u^{1/2}$, and Lemma 2 gives, uniformly in the center of Q,

$$\mathrm{MO}_Q(T(1)) + \mathrm{MO}_Q(T^*(1)) \lesssim \omega(u^{1/2})^{\alpha} + A(u)^{-\delta} \longrightarrow 0.$$

Thus the small-cube quantities Γ_1 vanish.

Large cubes. For each cube put $u = \ell(Q)$ and set

$$A(u) = \min\{u^{1/2}, \omega(u)^{-1/(d+1)}\}.$$

The functions ω and A are respectively decreasing and increasing. Also $A(u) \to \infty$, and $A\ell \geq u$, so for every threshold U the same estimate gives

$$\sup_{\ell(Q)\geq U} \big(\mathrm{MO}_Q(T(1)) + \mathrm{MO}_Q(T^*(1))\big) \lesssim \omega(U)^{\alpha} + A(U)^{-\delta} \longrightarrow 0.$$

Thus $\Gamma_2 = 0$.

Translations to infinity. Fix a cube Q_0. For $Q = Q_0 + x$ and $u = |x|$ sufficiently large, use the choice

$$A(u) = \min\{u^{1/2}, \omega(u/2)^{-1/(d+1)}\}.$$

Its center has magnitude at least $u/2$, so

$$\mathrm{MO}_{Q_0+x}(T(1)) + \mathrm{MO}_{Q_0+x}(T^*(1)) \lesssim \omega(u/2)^{\alpha} + A(u)^{-\delta} \longrightarrow 0$$

and every far-translation quantity $\Gamma_3^{Q_0}$ vanishes.

Condition (i) implies the bounded Stein bump estimate, so the source proof already gives $T(1), T^*(1) \in \mathrm{BMO}$ [1, Section 4]. The three limits above and the quoted characterization therefore yield

$$T(1), T^*(1) \in \mathrm{CMO}. \tag{4}$$

5 Completion of the proof

Necessity of (1) for compact T is established before Theorem 1 in [1]: normalized L^2 bumps converge weakly to zero in the extreme position/scale regimes, and compact operators map weakly convergent sequences to norm-convergent sequences.

Conversely, assume (1). The source proves directly by Cauchy–Schwarz that this condition implies the weak compactness property [1, Section 4]. Equation (4) supplies the other hypothesis in the compact $T(1)$ theorem quoted there from Mitkovski–Stockdale. Consequently T is compact on L^2. This proves Theorem 1.

6 Verification and novelty notes

The principal review points are:

1. the local distributional identity (3), including its harmless additive constant;
2. the factor $A^{d/2}$ in (2);
3. the slow-dilation choices in the three CMO regimes;
4. the exact direction in which the source compact $T(1)$ theorem is applied.

No numerical experiment is used: the proof is analytic.

A bounded official-arXiv search through 11 August 2026 used arXiv id 2405.08416, the exact title, and combinations of "L^∞ condition", "removed", "normalized bump", and "CMO". It found the source and adjacent compact $T(1)$ literature, but no later paper removing condition (ii). This packet settles only the linear theorem under the source's standard Calderón–Zygmund hypotheses; the bilinear analogue in its Remark 3.7 is not addressed. Expert review should focus first on Lemma 2 and the uniform tail modulus Ω.

References

[1] Á. Bényi, G. Li, T. Oh, and R. H. Torres, *Compact $T(1)$ theorem à la Stein*, J. Funct. Anal.; arXiv:2405.08416v2.

[2] M. Mitkovski and C. B. Stockdale, *On the $T1$ theorem for compactness of Calderón–Zygmund operators*, arXiv:2309.15819.